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END SEMESTER EXAMINATION-

2025

Name of the Course: M. Tech

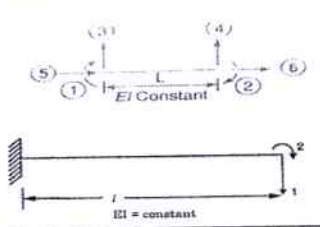
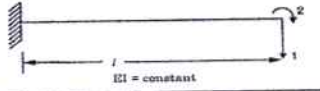
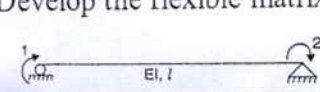
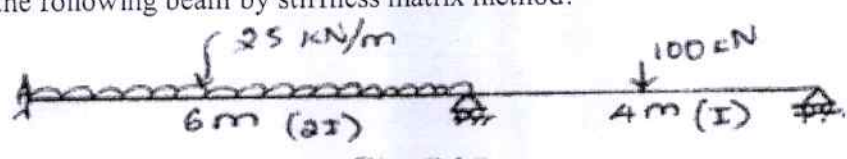
Semester : 1st semester

Name of the Paper : Matrix Method for Structural Analysis Paper Code: MSE 104

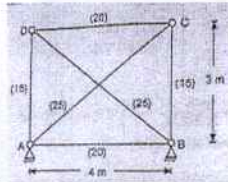
Time: 3 Hours

Maximum Marks: 100

- Note:
- All questions are compulsory
 - Answer any two questions from each sub question.
 - Each question carries 10 Marks

Q.1	Solve any Two (20 Marks)	
a)	Generate stiffness matrix for the following beam.  	(CO 1)
b)	Develop the flexible matrix for following 	(CO 1)
c)	(i) What are the differences between static and kinematic indeterminacies? Explain them with examples. (ii) Differentiate between DOF and DOR. (iii) Derive the relation between flexible and stiffness matrix	(CO 1)
Q.2	Solve any Two (20 Marks)	
a)	Analyze the following beam by stiffness matrix method. 	(CO 2)
b)	Analyze the following beam by stiffness matrix method. 	(CO 2)
c)	Analyze the continuous beam by stiffness matrix method	(CO 2)

		50 kN 15 kN/m 20 kN $E = 200 \times 10^6 \text{ N/mm}^2$ $I = 100 \times 10^3 \text{ mm}^4$	
Q.3	Solve any Two	(20 Marks)	
a)	Analyze the following beam by Flexibility Matrix Method	(CO 3)	
b)	Analyze the following frame by Flexibility Matrix Method	(CO 3)	
c)	Analyze the following beam by Flexibility Matrix Method	(CO 3)	
Q.4	Solve any Two	(20 Marks)	
a)	Write a Matlab program for force vector.	(CO 4)	
b)	Analyze the truss supported as shown in Fig, if the member AB is short by 5 mm. Take $E = 200 \text{ GPa}$. Cross sectional area of each member in cm^2 is indicated in brackets.	(CO 4)	
c)	Analyze the truss supported as shown in Fig. If the member AD is subjected to temperature drop of 3000°C . Take $E = 200 \text{ GPa}$ and coefficient of thermal expansion is 1.1×10^{-5} per degree centigrade. Cross sectional area of each member in cm^2 is indicated in brackets.	(CO 4)	



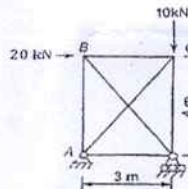
Q.5

Solve any Two

(20 Marks)

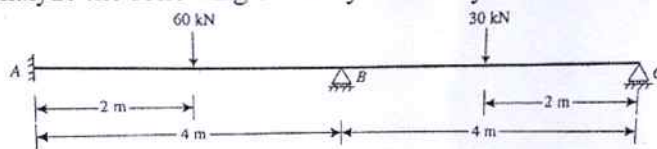
- a) Using the stiffness method, analyse the pin-jointed frame in figure. the cross sectional areas A and E are same for all the members.

(CO 5)



- b) Analyze the following beam by flexibility matrix method

(CO 5)



- c) Analyze the following frame by stiffness matrix method

(CO 5)

